Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 | www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

17 October 2026

Hiring Team
Perplexity

Application: Machine Learning Engineer (Search Ranking)

Dear Perplexity hiring team,

I am applying for the Machine Learning Engineer role on search quality. For three years at HDFC Asset Management I took retrieval and GenAI systems from prototype to production for people who used them every day, owning each one from the data and the model choices through to serving and release. I would like to bring that ownership to the ranking problems in your role, and I want to be direct that ranking at scale is the area where I have the most to learn.

At HDFC I designed and built the applications end to end: a ReactJS frontend, Python FastAPI microservices and retrieval with LlamaIndex, using Azure OpenAI embeddings and a separate pgvector database per department for semantic search over internal documents, with role-based access through the company LDAP server. I treated quality, cost and reliability as trade-offs: I self-hosted Llama-3 on a GPU EC2 instance for the largest department and kept the stronger Azure OpenAI outputs for decision makers. I packaged the services as Docker containers, deployed them on AWS through a one-click Jenkins and GitLab CI/CD pipeline, and ran a daily batch pipeline on SageMaker that parses over 300 rating-agency emails, fuzzy-matches company names and saves analysts 60 hours a month.

The closest thing to ranking in my background is my MSc dissertation with AstraZeneca. I co-built CellLineSelector, which ranks cancer cell lines by target inclusion and exclusion genes, and my focus was the data preprocessing, the scoring and ranking algorithm and the RAG layer: per-gene scores across six evidence layers, calibrated so that scores stay comparable across lineages and grouped into confidence tiers, refined with feedback from AstraZeneca scientists. In a group project I also compared CNN, ViT and ResNet encoders with symbolic, SBERT embedding-retrieval and LSTM decoders for image captioning, scored with SBERT cosine similarity, BLEU and ROUGE-L.

On your list, I have not owned a large-scale production ranking system, and I have not worked on search or recommender systems in production. I have not trained learning-to-rank models or neural rerankers, I have not worked with ranking metrics such as recall@k or NDCG or run online experiments, and I have not built low-latency multi-stage serving or feature computation. At HDFC quality came from user feedback, system-prompt guardrails and Compliance review rather than gold datasets or LLM-as-judge, and I have about three years of experience. The neural ranking side and the evaluation side are where I would expect to grow fastest, on a team that runs search at your scale.

What I can offer is the habit of owning a problem without it being decomposed for me. The first GenAI systems at HDFC were a two-person effort, my manager and me, and I worked directly with Client Service, Compliance and Investment to turn requirements into working tools; in my final year I acted as de facto team lead for up to four interns. Python is my day-to-day language and I am comfortable across the full lifecycle, from data pipeline to deployment to troubleshooting.

I am currently based in Bristol, hold a Student Visa and am eligible for the Graduate Visa, which gives two years of full working rights. Thank you for considering my application; I would be glad to talk it through.

Yours sincerely,
Kaustubh Sonawane